

Written examination in the course

Optimisation Theory
and Reliability

Monday June 11th 2018

kl. 9 - 13 (4 hours)

All usual helping aids are allowed, i.e. books, notes, calculator, computer etc. All communication equipment and computer communication protocols must be turned off.

The questions should be answered in English.

REMEMBER to write your study number and page number on all sheets handed in.

The set consists of nine exercises. The total weighting for each of the exercises is stated in percentage. You need 50 % in order to pass the exam.

It should be clear from the solution, which methods are used, and there should be a sufficient number of intermediate calculations, so the line of thought is clear.

Exercise 1: (10 %)

The following optimisation problem is considered:

$$\begin{aligned} \text{Minimise} \quad & f(\mathbf{x}) = (x_1 + 2)^2 + (x_2 + 2)^2 \\ \text{Subject to} \quad & g(\mathbf{x}) = 2 - x_1 - x_2 \leq 0 \end{aligned} \tag{1}$$

- a) Find the point(s) satisfying the KKT necessary conditions.
 b) Check if the point(s) is an optimum point using the graphical method (make a simple sketch).

Exercise 2: (15 %)

We will consider gradient-based minimisation of the following unconstrained function:

$$f(\mathbf{x}) = 5(x_1^2 - x_2) + (x_2 - 1)^2 - 2x_1 + 3 \tag{2}$$

The starting point is: $\mathbf{x}^{(0)} = [3, 4]^T$.

- a) Complete the first iteration of the steepest descent method for the function. The 1D line search problem should be solved analytically.
 b) Complete the first iteration of the modified Newton's method for the function. The 1D line search problem should be solved analytically.

Exercise 3: (10 %)

An optimisation problem is formulated as:

$$\text{minimise} \quad f(\mathbf{x}) = 10x_1^4 + 2x_2^2 - 2x_1^2x_2 - 5x_1 - x_2$$

Subject to:

$$\begin{aligned} g_1(\mathbf{x}) &= x_2 - 10 && \leq 0 \\ g_2(\mathbf{x}) &= 2x_1 + 4x_2 - 10 && \leq 0 \\ g_3(\mathbf{x}) &= x_1 - x_2 - 1 && \leq 0 \end{aligned}$$

The initial starting point for an optimisation is $\mathbf{x}^{(0)} = [0, 1]^T$, for which a search direction is determined to $\mathbf{d}^{(0)} = [1.5, 0.5]^T$ and the vector of Lagrange multipliers for the constraints is $\mathbf{u} = [0, 0, 3.5]^T$. Let $R_0 = 1$ and $\gamma = 0.5$. Choose the trial step according to the sequence $t_0 = 1, t_1 = \frac{1}{2}, t_2 = \frac{1}{4}, t_3 = \frac{1}{8} \dots$

Calculate the step size using the inexact line search procedure (approximate step size procedure) and determine the new design variables, $\mathbf{x}^{(1)}$, for the next iteration.

Exercise 4: (7 %)

The following multi-objective optimisation problem is considered:

$$\begin{aligned} \text{minimise} \quad & f_1(\mathbf{x}) = (x_1 - 2)^2 + (x_2 - 8)^2 + 6 \\ & f_2(\mathbf{x}) = (x_1 - 10)^2 + (x_2 - 4)^2 + 2 \end{aligned}$$

Figure 1 shows respectively the contour curves for the objective functions including the Pareto optimal points (left) and the Pareto front in the criterion space (right).

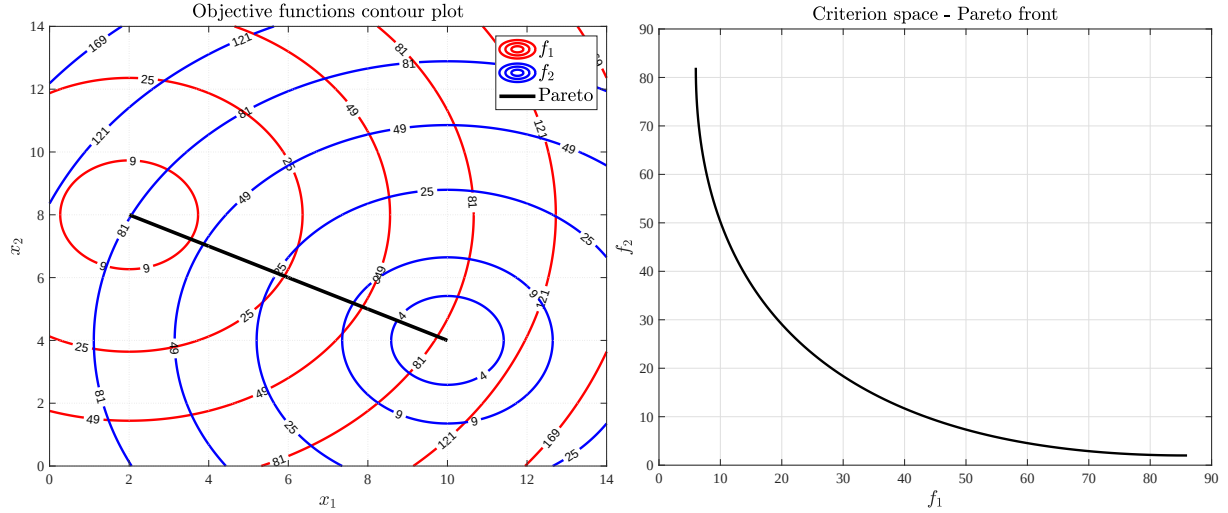


Figure 1: Left: Contour curves and Pareto optimal points. Right: The Pareto front in the criterion space

- Determine the values of the Utopia point.
- Assume that the multi-objective problem is solved as single objective problem, $U(\mathbf{x})$, using the weighting method with $w_1 = 2$ and $w_2 = 1$. Determine analytically the minimum objective function value $U(\mathbf{x}^*)$, and the optimum set of design variables $\mathbf{x}^*$.

Exercise 5: (8 %)

Solve the following problem by setting up the solution tree and using the *Local Minimization Branch & Bound Method*:

$$\text{minimise} \quad f(\mathbf{x}) = 2x_1 - 5x_2$$

Subject to:

$$g_1(\mathbf{x}) = -0.5x_1 + x_2 - 4.7 \leq 0$$

$$g_2(\mathbf{x}) = 2x_1 + x_2 - 16 \leq 0$$

$$g_3(\mathbf{x}) = x_1 - x_2 - 2 \leq 0$$

Both x_1 and x_2 should be integer values

As a help for solving the problem the objective function contours and constraints are plotted in figure 2.

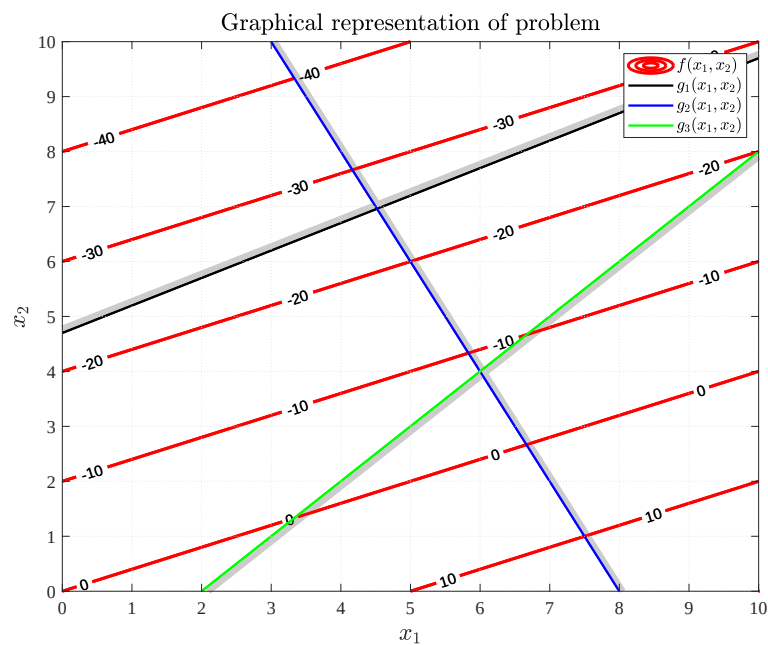


Figure 2: Graphical representation of problem in exercise 5.

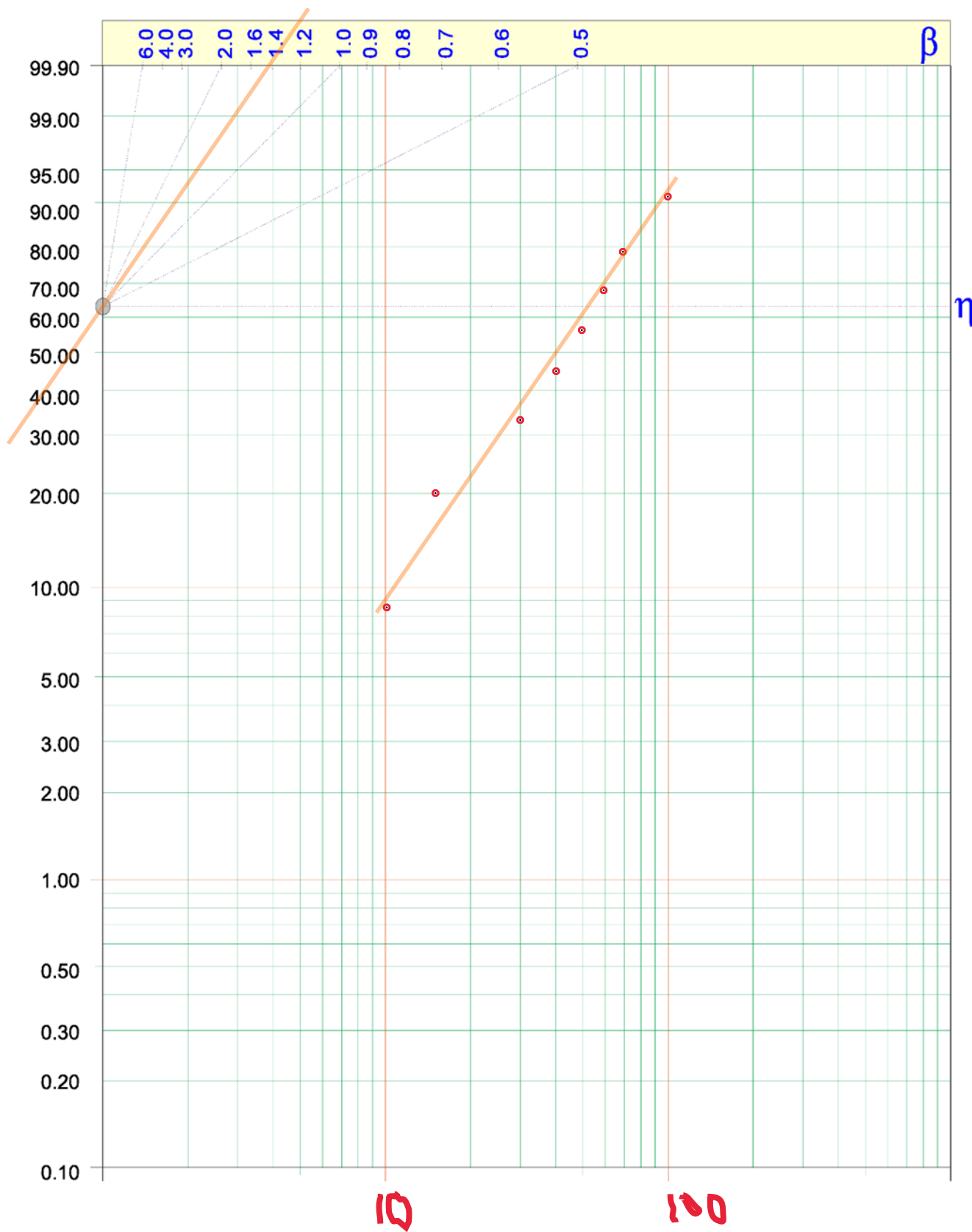
Question 6 (14%)

A group of 8 paper clips (no. 1, no. 2, ..., to no. 8) are tested to study the wear out behavior with a bending angle of 90° . The cycles to failure for clips no. 1 to no. 8 are recorded as: 15, 50, 40, 10, 30, 70, 100, and 60. Another group of 8 paper clips (no. 9, no. 10, ..., to no. 16) are tested with a bending angle of 180° . The cycles to failure for clips no. 9 to no. 16 are recorded as: 25, 10, 4, 12, 15, 7, 18, and 2.

- 1) (3%) Arrange these time-to-failure numbers by using median ranking method for each group of the tested paper clips .
- 2) (5%) Plot the cycles to failure of each group of the samples in the attached Weibull paper on next page. Please find out the values for θ and η for each group of testing results, respectively.
- 3) (3%) Find the B10 lifetime (the cycle when unreliability is equal to 0.1 or 10%) of the paper clips with the bending angle of 90° and 180° , respectively.
- 4) (3%) Assume that the relationship between the cycle to failure and the bending angle follows a power law, please find the relationship between B10 lifetime (with the results based on median ranking above) of the paper clips and bending angle, based on the results from 3).

Appendix I – Median rank table

sample size = n failure rank = i										
n										
i	1	2	3	4	5	6	7	8	9	10
1	.5000	.2929	.2063	.1591	.1294	.1091	.0943	.0830	.0741	.0670
2		.7071	.5000	.3864	.3147	.2655	.2295	.2021	.1806	.1632
3			.7937	.6136	.5000	.4218	.3648	.3213	.2871	.2594
4				.8409	.6853	.5782	.5000	.4404	.3935	.3557
5					.8706	.7345	.6352	.5596	.5000	.4519
6						.8906	.7705	.6787	.6065	.5481
7							.9057	.7979	.7129	.6443
8								.9170	.8194	.7406
9									.9259	.8368
10										.9330



Question 7 (11%)

A wooden pencil consists of four sub-systems of wooden shaft, lead shaft, eraser, and ferrule, as shown below.



In order to prepare the information used for Failure Mode and Effect Analysis (FMEA), please answer the following questions. The answers of the questions will be evaluated mainly on the understanding of the procedures and NOT on the accuracy of the wording of specific descriptions.

- 1) (3%) Assume the “wooden pencil” is the system of interest, plot its boundary diagram to show at least two interfaces of it.
- 2) (4%) Plot the Parameter diagram (P-diagram) of one of the sub-systems (choose one only, either wooden shaft, lead shaft, eraser, or ferrule), to include the main aspects of a P-diagram (i.e., input signals, ideal response, noise factors, control factors, and error states).
- 3) (4%) Identify two failure modes and two failure causes of the sub-system you chose in 2) based on the plotted P-diagram.

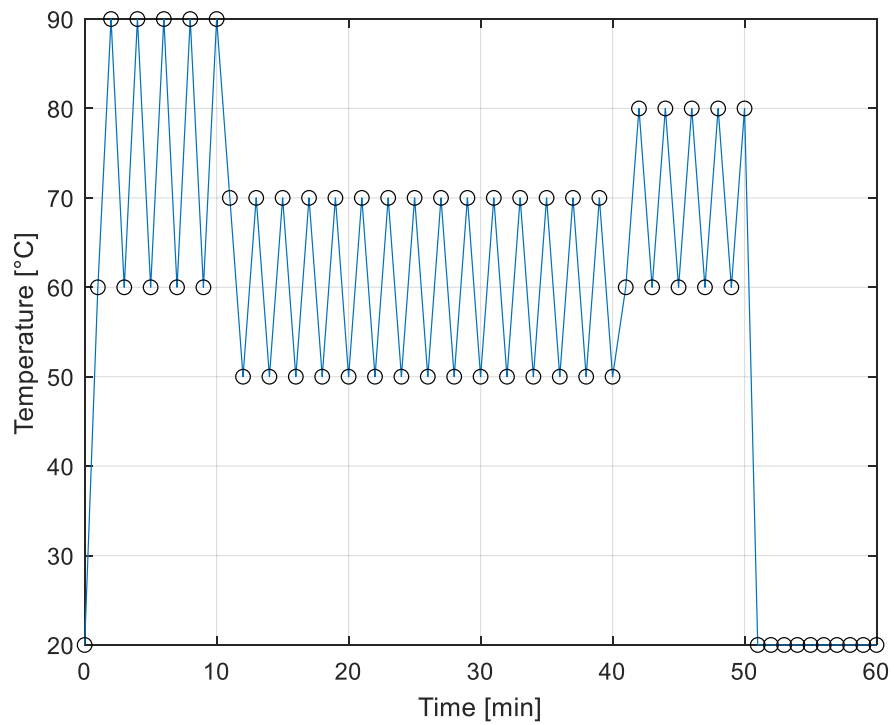
Question 8 (12%)

A system comprises two components from two different manufacturers M1 and M2. For a correct operation, both components have to work. Manufacturer M1 provides a Weibull reliability function with $\beta_1 = 1.6$ and $\eta_1 = 75000$ h. Manufacturer M2 provides an exponentially distributed reliability function with $MTBF_2 = 100000$ h.

- 1) (2%) Sketch up an RBD schematic of the overall system;
- 2) (5%) Calculate the reliability of each component after 1 year;
- 3) (3%) Work out the overall reliability function, and the overall system reliability after 1 year.
- 4) (2%) Calculate or give a good estimation of the overall system B_{10} lifetime.

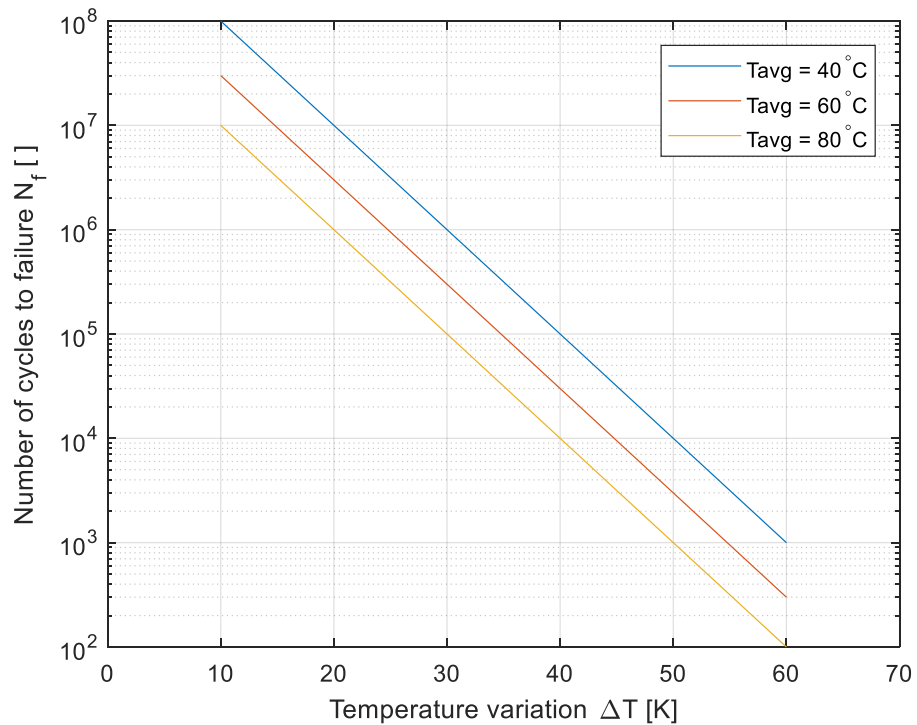
Question 9 (13%)

The 1-hour mission profile of a milling head is reported in the following figure, where the temperature is plotted against time. The same profile is repeated 10 hours per day, 7 days per week.



1-hour mission profile of a milling head

The milling head lifetime has been evaluated and modeled as Coffin-Manson lifetime model, i.e. taking into account temperature average (T_{avg}) and variation (ΔT), as shown in the following chart.



Coffin-Manson chart of milling head

Questions:

- 1) (2%) Identify and list in a table the three different stress levels (T_{avg} , ΔT) occurring in the given mission profile;
- 2) (3%) Identify such levels as points in the Coffin-Manson chart and complete the above table with the number of cycles to failure N_f for each of them;
- 3) (6%) Calculate the accumulated damage (in %) within 1 hour;
- 4) (2%) Calculate the expected lifetime.